Conceptual Model Selection Analysis — EMS Readiness Optimization

> ⚠️ **Note:** This document references historical P0 metrics (e.g., 8.08 min) in narrative context to explain the rationale for the nomenclature migration (DEC-011/DEC-012). The **current** P0 baseline is spatially-stratified (mean RT ≈ 3.17 min at K=20). See `nomenclature_migration.md` (nomenclature_migration.md) for details.
>
> **Date:** March 15, 2026
> **Purpose:** Comprehensive review of the conceptual model landscape, what was implemented, and the rationale behind selection decisions.

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1. Full Conceptual Landscape

The project addresses a **facility location / resource allocation** problem: given K ambulance units and 48 candidate FDNY firehouses in Manhattan, determine the allocation that minimises response time and maximises coverage. The conceptual model document (`docs/conceptual_model.md`) and decisions log identify the following approaches that were **considered** during the project design:

1.1 Optimization Models Considered

#	Approach	Category	Status
1	**Demand-Weighted Allocation (P2)**	MIP — minimise demand-weighted response time	Yes Implemented & primary recommendation
2	**P-Median (P2-alt / P2b)**	MIP — minimise total (demand-weighted) distance	Yes Implemented
3	**Maximal Coverage (P2-cov / P2c)**	MIP — maximise demand covered within τ-minute threshold	Yes Implemented
4	**Set Covering**	MIP — cover all demand with minimum units	No Not implemented (subsumed by models 1–3)
5	**Multi-Objective / Pareto Optimization**	Bi-objective: response time vs. equity	No Deferred to future work
6	**Time-Varying Optimization (P3)**	Hour-specific λ tables, dynamic reallocation	No Deferred to future work
7	**CBD-Focused Demand-Weighted**	MIP — minimise response time for CBD precincts only	Yes Implemented (robustness check)
8	**CBD-Focused Maximal Coverage**	MIP — maximise CBD coverage within threshold	Yes Implemented (robustness check)
9	**Metaheuristics (GA, SA, etc.)**	Flexible but tuning required	No Not implemented (MIP solves fast enough)
10	**Heuristic approaches**	Fast but no optimality guarantee	No Not implemented as optimization (but baselines serve this role)

1.2 Baseline Policies Considered

#	Approach	Category	Status
11	**Uniform Allocation (P0)**	Non-optimized — equal distribution	Yes Implemented
12	**Demand-Proportional Allocation (P1)**	Non-optimized — units proportional to nearest demand	Yes Implemented
13	**Spatially-Stratified P0 (P0 (spatially-stratified), latitude)**	Non-optimized — even geographic coverage	Yes Implemented (replaced original P0 in current version)
14	**Spatially-Stratified P0 (grid)**	Non-optimized — grid-based firehouse selection	Yes Implemented
15	**Spatially-Stratified P0 (maximin)**	Non-optimized — greedy farthest-point heuristic	Yes Implemented
16	**Random allocation**	Non-optimized — random with seed	No Not implemented (non-reproducibility concerns)

1.3 Simulation Approaches Considered

#	Approach	Status
17	**Discrete-Event Simulation (DES) via SimPy**	Yes Implemented
18	**Agent-Based Simulation**	No Rejected (too complex for this application)
19	**System Dynamics**	No Rejected (too aggregated)
20	**Analytical Queueing Models**	No Rejected (spatial component not representable)

1.4 Distance Metrics Considered

#	Metric	Status
21	**Haversine (great-circle)**	Yes Implemented (primary)
22	**Manhattan (taxicab) distance**	Yes Implemented (robustness check)
23	**Road network distance (OSRM/Google)**	No Not implemented (out of scope)

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2. Models Selected for Implementation

2.1 Core Optimization Models (in `models.py`)

Function	Policy Code	Objective
`build_demand_weighted()`	P2	Minimise demand-weighted response time
`build_p_median()`	P2-alt	Minimise total demand-weighted distance (capacity-aware variant)
`build_maximal_coverage()`	P2-cov	Maximise demand-weighted coverage within 8-min threshold
`build_cbd_focused_demand_weighted()`	—	Minimise CBD-only demand-weighted response time
`build_cbd_focused_coverage()`	—	Maximise CBD-only coverage

Total: 5 optimization formulations implemented as PuLP MIP models.

2.2 Baseline Policies (in `policies.py`)

Function	Policy Code	Method
`uniform_allocation()`	P0 (original)	Equal distribution, round-robin remainder
`demand_proportional_allocation()`	P1	Units proportional to nearest-firehouse demand credit
`spatially_stratified_allocation()`	P0 (spatially-stratified)	Three sub-methods: latitude, grid, maximin

Total: 3 baseline allocation functions (with P0 (spatially-stratified) offering 3 sub-variants).

2.3 Mapping to Conceptual Framework

The conceptual model document (§1.1) defines **5 candidate policies** for simulation comparison:

Code	Policy	Implementation
P0	Uniform allocation	`uniform_allocation()` → replaced by `spatially_stratified_allocation(method='latitude')` in current version
P1	Demand-proportional	`demand_proportional_allocation()`
P2	Demand-weighted optimisation	`build_demand_weighted()`
P2-alt	P-median selection	`build_p_median()`
P2-cov	Maximal coverage	`build_maximal_coverage()`

The CBD-focused models and Manhattan distance variants were **additional robustness checks** beyond the core 5 policies.

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3. Selection Rationale

3.1 Why These Three MIP Models?

The three core optimization models were chosen to represent **distinct operational objectives** (DEC-003):

1. **Demand-Weighted (P2):** Aligns with equity — weights response time by demand intensity, so high-demand areas receive proportionally better service. This is the **primary recommendation**.

2. **P-Median (P2-alt):** Answers “which K locations should we open?” — useful when the decision is about which firehouses to staff rather than how many units to place.

3. **Maximal Coverage (P2-cov):** Focuses on a binary service-level target (≤ 8 min) — useful for regulatory compliance and equity guarantees.

These three represent the **canonical facility location models** in the operations research literature (Daskin 2013; Church & ReVelle 1974; ReVelle & Swain 1970).

3.2 Why Were Other Models Excluded?

Excluded Model	Reason
Set Covering	Subsumed by Maximal Coverage — Set Covering asks “what’s the minimum K to cover everything?” while Maximal Coverage asks “given K, what’s the best coverage?” The latter is more relevant for fixed-budget decisions.
Multi-Objective Pareto	Adds significant complexity; deferred to future work. The single-objective models already allow comparing equity (P2 vs. P2-cov) indirectly.
Time-Varying (P3)	Requires dynamic reallocation during the simulation, which is out of scope for this **static staging** study. Listed as future work.
Metaheuristics	Unnecessary — the problem size (48 firehouses × 25–30 precincts) is small enough that CBC solves all MIPs in under 5 seconds. No need for approximate methods.
Road Network Routing	Requires external API integration (OSRM/Google) and adds complexity without changing relative policy rankings (confirmed by Manhattan distance robustness check).

3.3 Why Two Baseline Policies?

- **P0 (Uniform):** A naïve lower bound — distributes units without any demand information. Answers "how much does intelligence in allocation actually matter?"
- **P1 (Demand-Proportional):** A practical heuristic requiring no solver — answers "can a simple rule-of-thumb approach the optimal?"

The **P0 (spatially-stratified)** upgrade (DEC-011) was motivated by discovering that the original P0 was artificially weak due to database-order bias, not a genuine "uniform" allocation. The latitude-based stratification corrected this, providing a fairer baseline (mean RT dropped from 8.08 → 3.17 min at K=20).

3.4 Why CBD-Focused Models?

The CBD accounts for ~56% of total demand. CBD-focused models (DEC-009) were implemented to test whether **geographic prioritisation** could improve performance in the highest-demand zone. The result was definitive: CBD-focused optimization **does not** improve CBD response time (+1.2%) but **sharply degrades** non-CBD service (+159% RT). This validates that the Manhattan-wide P2 already handles the CBD well.

3.5 Why Manhattan Distance as Robustness Check?

Manhattan distance (DEC-008) was implemented because Haversine underestimates road distances in a grid network. The side-by-side comparison showed **near-identical simulation results** (mean RT 2.55 min for both), with only 2 of 48 firehouses differing between allocations. This confirms Haversine is a sufficient approximation.

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4. Experimental Coverage

4.1 Core Experiments (from `experimental_design.md`)

The formal experimental design uses **3 policies × 4 experiments = 1,440 total simulation runs**:

Experiment	Policies	Factor Varied	Runs
Exp 1: Policy Comparison	P0, P1, P2	Baseline conditions (K=20)	90
Exp 2: Fleet Sensitivity	P0, P1, P2	K ∈	540
Exp 3: Demand Sensitivity	P0, P1, P2	δ ∈	540
Exp 4: Service Robustness	P0, P1, P2	μ_s ∈ {20, 25, 30} min	270

Note: The experimental design focuses on the 3 main policies (P0, P1, P2). The P2-alt and P2-cov models are evaluated in the optimization phase (Phase 3) but are **not** carried through all 4 simulation experiments — this is a deliberate scoping decision to keep the experiment tractable.

4.2 Additional Robustness Analyses

Beyond the core experiments, the following were conducted:

Analysis	Models Used	Purpose
Capacity sensitivity	P0, P1, P2 across cap ∈	Determine default capacity (→ cap=2)
Manhattan vs. Haversine distance	P2 with both metrics	Validate distance metric robustness
CBD-focused optimization	CBD-DW, CBD-Coverage vs. P2	Test geographic prioritisation
Spatial stratification comparison	P0-latitude, P0-grid, P0-maximin	Determine best P0 variant

4.3 Coverage Assessment

Dimension	Coverage
Policy types	5/5 conceptual policies implemented (P0, P1, P2, P2-alt, P2-cov)
Fleet sizes	9 values tested:
Demand variation	6 multipliers:
Service time variation	3 levels: {20, 25, 30} minutes
Distance metrics	2: Haversine (primary) + Manhattan (robustness)
Capacity constraints	5 levels:
Geographic focus	2: Manhattan-wide + CBD-focused

The experimental coverage is comprehensive — all core conceptual models are implemented and tested, with extensive sensitivity analysis. The only deferred items (multi-objective Pareto, time-varying P3, road network routing) are clearly documented as future work.

5. Summary: How Model Selection Supports Research Goals

Research Questions → Model Mapping

Research Question	Models Used	Finding
RQ1: How do policies compare on response time and coverage?	P0, P1, P2 (+ P2-alt, P2-cov)	P2 dominates: 2.49–2.54 min mean RT, 100% 8-min coverage
RQ2: How sensitive is performance to fleet size K?	P0, P1, P2 across K ∈	Diminishing returns beyond K=30; P2 advantage narrows with more units
RQ3: How robust is performance to demand changes?	P0, P1, P2 across δ ∈	P0 degrades fastest under high demand (H3 confirmed)
RQ4: How do service time variations affect policy performance?	P0, P1, P2 across μ_s ∈	P2 most robust to service time variation (H4 confirmed)

Key Methodological Strengths

1. **Canonical models:** The three MIP formulations (demand-weighted, p-median, maximal coverage) represent the standard operations research toolkit for facility location problems.
2. **Fair baselines:** The P0 (spatially-stratified) upgrade ensures the baseline is a genuine geographic benchmark, not an artifact of database ordering.
3. **Robustness via multiple dimensions:** Distance metric, capacity, demand intensity, service time, and geographic focus are all tested.
4. **Practical scoping:** Excluded models (metaheuristics, road network, dynamic reallocation) are justified and documented rather than silently omitted.
5. **Simulation validation:** All optimization solutions are evaluated through DES, not just static objective values — this captures queueing dynamics that MIP alone cannot.

Conclusion

The project considered **~20+ distinct conceptual approaches** across optimization models, baseline policies, simulation methodologies, and distance metrics. Of these, **5 optimization formulations + 5 baseline policies + 1 simulation framework + 2 distance metrics** were implemented — covering the core conceptual landscape comprehensively. Selection decisions are well-documented in the decisions log, with clear rationale for both inclusions and exclusions. The experimental design provides thorough coverage of the parameter space relevant to the 4 research questions.

Analysis prepared: March 15, 2026
Sources: `docs/conceptual_model.md`, `docs/decisions_log.md`, `docs/experimental_design.md`, `docs/optimization_formulation.md`, `docs/assumptions_log.md`, `src/ems_readiness/optimization/models.py`, `src/ems_readiness/optimization/policies.py`